

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

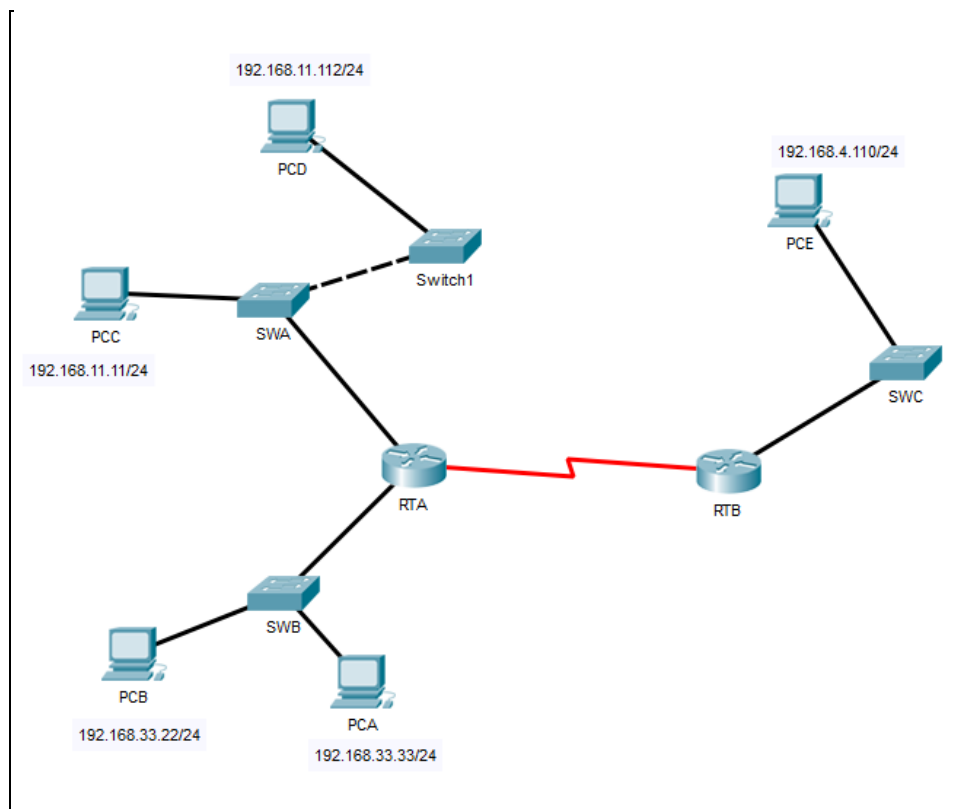


Figure 1

Part 1: Review the topology

Step 1: Open the **Lab_4_ARP.pkt**, and Perform the following tasks.

- At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0002.4a00.0e91    DYNAMIC   Fa0/1
1       000c.8546.7d85    DYNAMIC   Fa1/1
SWA#
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       000c.cf0c.593a    DYNAMIC   Fa0/1
SWB#
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
SWC#
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

```
C:\>arp -a
No ARP Entries Found
C:\>
```

- e. What are your thoughts on the results?
No ARP means that PCA haven't send any traffic.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- a. Click **PCA** and click the Desktop tab > Command Prompt.
b. Enter the **ping 192.168.33.22** command. This may take a few seconds.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time=1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

- c. In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746       dynamic

C:\>
```

- d. In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06        dynamic
C:\>
```

- e. In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.

PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- a. Click **PCC** and click the Desktop tab > Command Prompt.
- b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
C:\>
```

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
    Internet Address      Physical Address         Type
192.168.11.1             0002.4a00.0e91          dynamic
192.168.11.112           0001.6462.0278          dynamic

C:\>|
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Reply from 192.168.4.110: bytes=32 time=23ms TTL=126
Reply from 192.168.4.110: bytes=32 time=19ms TTL=126
Reply from 192.168.4.110: bytes=32 time=20ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 23ms, Average = 15ms

C:\>arp -a
    Internet Address      Physical Address         Type
192.168.11.1             0002.4a00.0e91          dynamic
192.168.11.112           0001.6462.0278          dynamic
```

- e. Discuss the results you got from all the commands on PCC.

First, when we ping PCB, the ARP table contains the router's MAC address 192.168.11.1, and does not have PCB's MAC address, indicating the communication happens through a router.

Then, when we ping PCD, PCD's MAC address added in the ARP table which means that PCD is in the same subnet as PCC.

Lastly, when we ping PCE, PCE's MAC address does not appear in the ARP table which means that the communication between PCC and PCE happens through a router.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
```

```

RTA#show arp

RTA>enable
RTA#show arp
Protocol    Address            Age (min)  Hardware Addr  Type   Interface
Internet    192.168.11.1       -          0002.4A00.0E91  ARPA   FastEthernet
Internet    192.168.11.11      39         00D0.D39A.C0D9  ARPA   FastEthernet
Internet    192.168.33.1       -          000C.CF0C.593A  ARPA   FastEthernet
Internet    192.168.33.22      39         0060.47EA.A746  ARPA   FastEthernet
RTA#

```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```

RTB>enable
RTB#show arp

RTB>enable
RTB#show arp
Protocol    Address            Age (min)  Hardware Addr  Type   Interface
Internet    192.168.4.1        -          0001.977A.B614  ARPA   FastEthernet
Internet    192.168.4.110      38         0060.702D.7C08  ARPA   FastEthernet
RTB#

```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```

SWA>enable
SWA#show arp

SWA#show mac-address-table

SWA>enable
SWA#show arp

SWA#show mac-address-table
          Mac Address Table
-----
Vlan      Mac Address      Type        Ports
----      -
1         0002.4a00.0e91   DYNAMIC     Fa0/1
1         000c.8546.7d85   DYNAMIC     Fa1/1
SWA#

```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       000c.cf0c.593a   DYNAMIC     Fa0/1
SWB#
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0001.977a.b614   DYNAMIC     Fa0/1
SWC#
```

```

Switch>enable
Switch#show arp

Switch#show mac-address-table
          Mac Address Table
-----
Vlan      Mac Address      Type      Ports
----      -
1         0005.5e95.187b    DYNAMIC   Fa0/1
Switch#

```

- d. Do switches use arp table? (Y/N)

Yes

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

when we send a ping from the switch say to its default gateway, if there is no MAC address for the default gateway is already in the switch's MAC address table, then the switch will ARP for the default gateway MAC address by sending an ARP request to the IP address of the default gateway with a Dest MAC address of FFFF.FFFF.FFFF.

The default gateway will then reply with its MAC address as the source MAC address and the destination MAC the switch's MAC address.

- f. What information does the command **show mac-address-table** gives?

MAC Address, Vlan, Type and Ports

Part 3: Attach wireless lab results.

In this part, you will use **Lab_4_Wireless.pka** file.

Step1: Change the filename of the pka file.

- Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The network topology shows a central switch S1 connected to a wireless router WRS2. S1 has three subinterfaces: G0/0.10 (VLAN 10), G0/0.20 (VLAN 20), and G0/0.88 (VLAN 88). PC1 is connected to S1 G0/0.10, PC2 to S1 G0/0.20, and PC3 to WRS2. WRS2 is connected to S1 F0/18 and has a wireless interface G0/1. The configuration window for WRS2 shows the following addressing table:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

The PT Activity window shows the completion status: 100/100.

- c. Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

The network topology is the same as the previous image. The configuration window for WRS2 shows the following addressing table:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

The PT Activity window shows the completion status: 100/100.